Discovering LTE and 5G

Mobile Networks

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https://github.com/kiarabartuccio/MobileNetworksProject.git
November 10, 2026

Hardware

We will be using our cellular devices and the built-in Field-test mode on the iphones.

1. IPhone 13:
    a. Plan: Telus plan (Family plan with 3 users with a total of 324.31 GB of high speed data)
2. IPhone 17 Pro Max:
    a. Plan: Telus plan (Family plan with 175 GB of high speed data, no data overages)

Type	Manufacturer	Model	Quantity	Owner	Status
Smartphone	IPhone	IPhone 13	1	Audréane Deschênes	Active
Smartphone	Phone	IPhone 17 Pro Max	1	Kiara Bartuccio	Active
Laptop	Lenovo	IdeaPad Slim 5 16ABR8	1	Audréane Deschênes	Active
Laptop	Dell	Inspiron 5575	1	Kiara Bartuccio	Active

Hosting

We did not need to rent a machine. No VPS(virtual private server) or public server is needed for this project. Measurements will be done on our two smartphones, and documentation will be done on the computers at Vanier college and our personal laptops. We use our smartphones to test and get LTE and 5G signals, while we use our laptops and school computers for research, documentations, slides, questions, and to access our GitHub repository.

Software

- Field-test Mode(hidden diagnostic feature on phone that lets you see technical information about cellular connection, like signal strength and the network you're connected to)

- To view cellular signal measurements like RSRP, closer to 0 means a better signal

    -> On iPhone:

    - Get the Field test mode shortcut (IPhone only): https://www.icloud.com/shortcuts/00a9f75d55b44f9c938aa71f021ae62d
    - Run the shortcut.
    - Read RSRP (Signal Strength) and RSRQ (Signal Quality).
    - In this shortcut, you can view both the LTE and 5G results at the same time (live).

- Discovered a location with weak or no cellular reception.
- Use Speedtest by Google to measure download speed, upload speed, and latency.
- Compare the results with a mobile carrier's coverage map app that was downloaded from the apple app store.
- Save screenshots and measurements in the evidence/ folder.

How to Read the Values:

1. RSRP (Signal Strength)

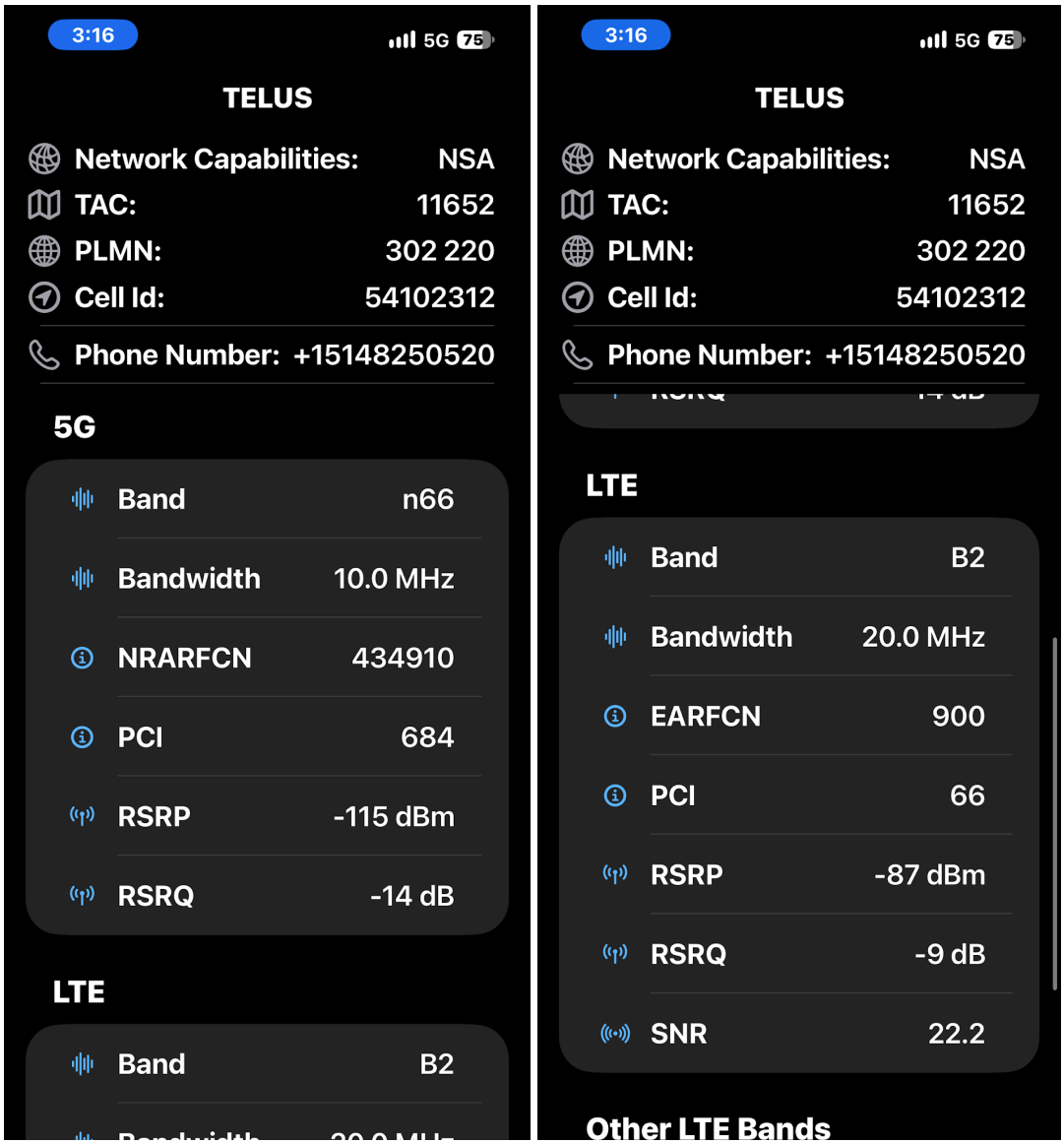
- ≥ -80 dBm: Excellent (maximum data speeds)
- -80 dBm to -90 dBm: Good (strong signal and speeds)
- -90 dBm to -100 dBm: Fair to poor (marginal data, possible drop-outs)
- < -100 dBm: Poor (severe performance drop, risk of disconnect)

2. RSRQ (Signal Quality)

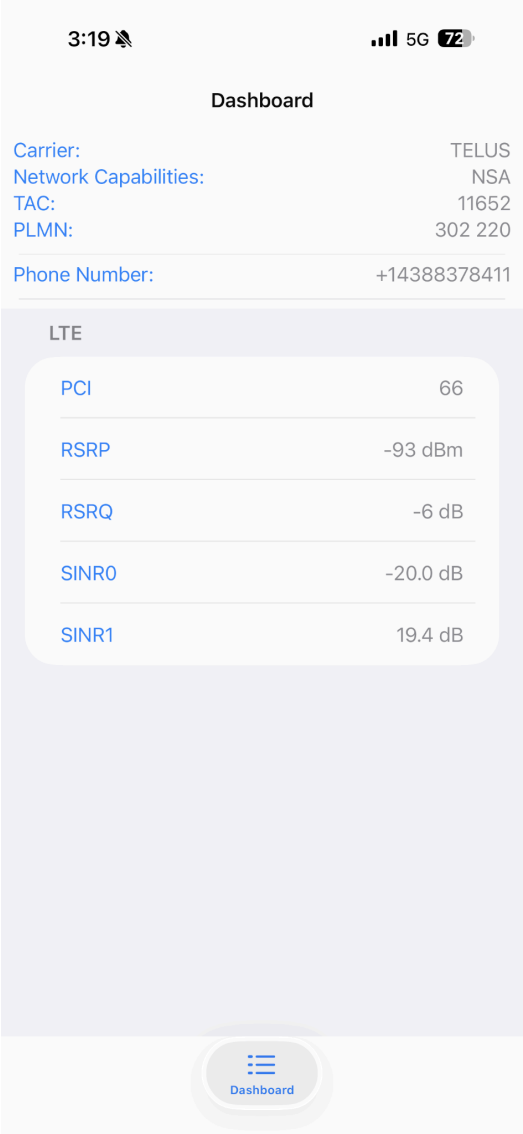
- ≥ -10 dB: Excellent (clean channel, max speeds)
- -10 dB to -15 dB: Good (normal operation)
- -15 dB to -20 dB: Fair to poor (high interference or load)
- ≤ -20 dB: Poor (heavy congestion or interference, frequent drop-outs)

In the lab class D-241:

Audreane:


3:16
5G 75
TELUS
Network Capabilities: NSA
TAC: 11652
PLMN: 302 220
Cell Id: 54102312
Phone Number: +15148250520
5G
Band n66
Bandwidth 10.0 MHz
NRARFCN 434910
PCI 684
RSRP -115 dBm
RSRQ -14 dB
LTE
Band B2
3:16
5G 75
TELUS
Network Capabilities: NSA
TAC: 11652
PLMN: 302 220
Cell Id: 54102312
Phone Number: +15148250520
LTE
Band B2
Bandwidth 20.0 MHz
EARFCN 900
PCI 66
RSRP -87 dBm
RSRQ -9 dB
SNR 22.2
Other LTE Bands

Kiara:

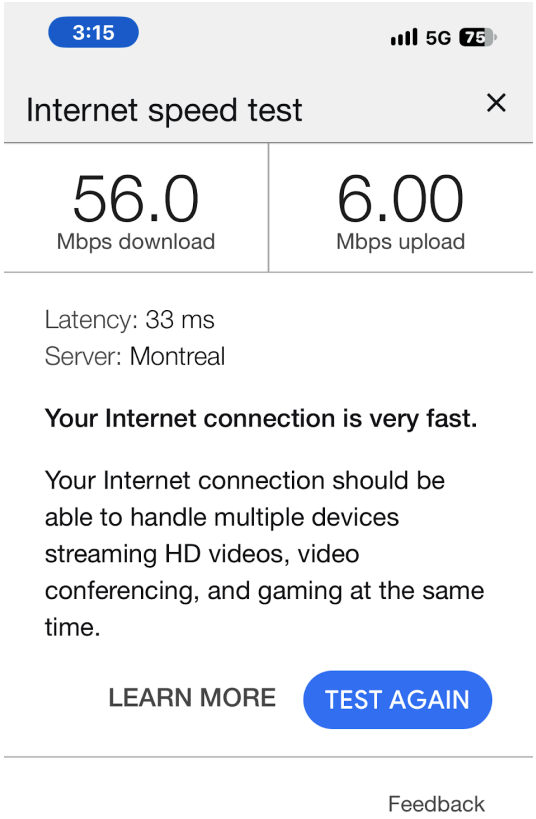


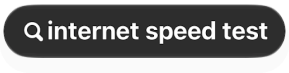
Kiara's notes: While testing, only LTE was available for me unlike Audreane which has LTE and 5G. I find it strange since at the top you can see I am connected to 5G. It could be because of the signal strength and network conditions which is why we will go around the school for our actual testing time. Maybe the shortcut just doesn't display 5G. My iphone is set to 5G Auto which uses 5G when appropriate and tried 5G On which prefers 5G when available

- Data measuring app called "Internet Speed Test Speedcheck" or a classic internet speed test through Google

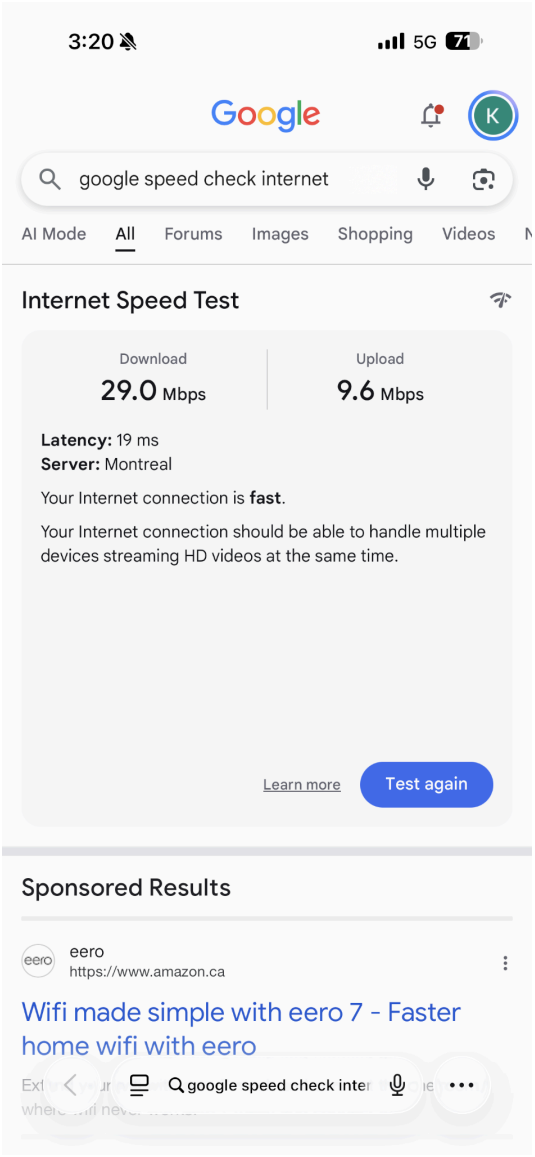
  - To measure download speed, upload speed, latency, and server

Audreane:


3:15
5G
Internet speed test
56.0
Mbps download
6.00
Mbps upload
Latency: 33 ms
Server: Montreal
Your Internet connection is very fast.
Your Internet connection should be able to handle multiple devices streaming HD videos, video conferencing, and gaming at the same time.
LEARN MORE
TEST AGAIN
Feedback


internet speed test

Kiara:

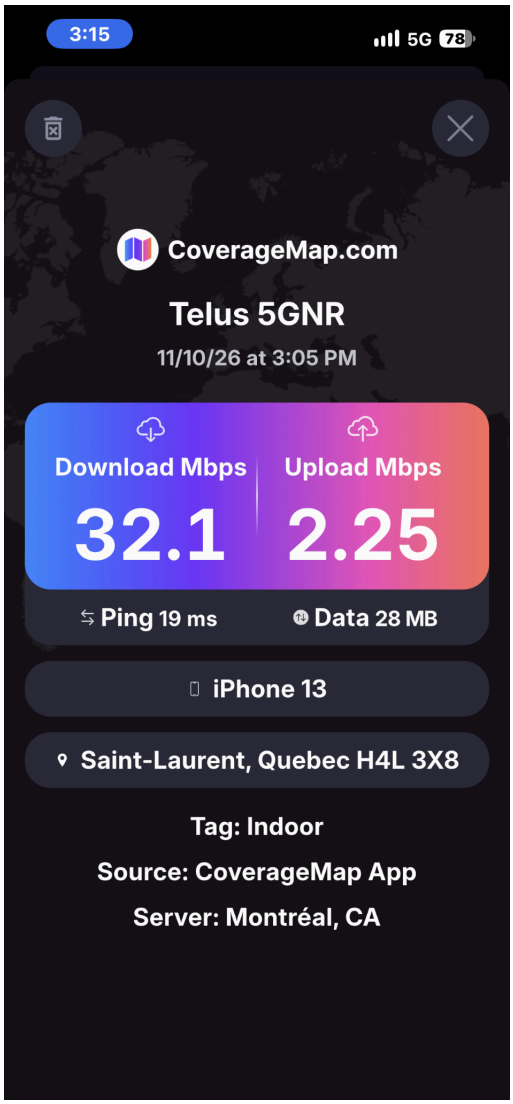


Kiara's Notes: Although my partner and I are right next to each other we can see a clear difference in data we both got. Audreane's is faster than mine.

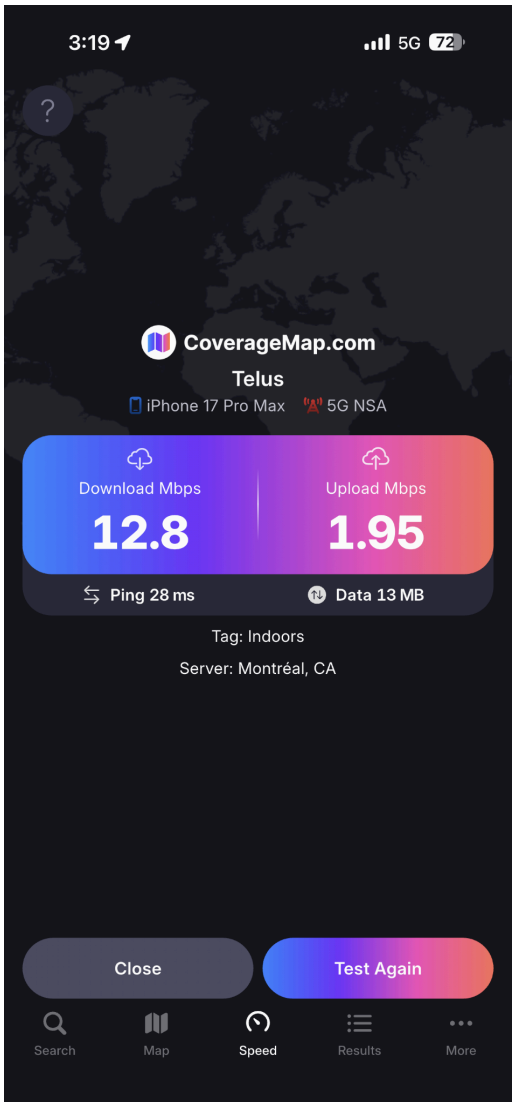
- Carrier coverage map(map that shows where a mobile carrier like Bell, Rogers, or TELUS provides cellular service, including LTE (4G) and 5G) - To check expected LTE and 5G coverage, compare LTE/5G coverage with our measurements, and explain any differences between the map and our actual reception.

  - We used Coverage Map app from the apple store

Audreane:


3:15
5G 78
CoverageMap.com
Telus 5GNR
11/10/26 at 3:05 PM
Download Mbps
32.1
Upload Mbps
2.25
Ping 19 ms
Data 28 MB
iPhone 13
Saint-Laurent, Quebec H4L 3X8
Tag: Indoor
Source: CoverageMap App
Server: Montréal, CA

Kiara:



Kiara's notes: Although Audreane are both indoors and close to each other, we received different data again.

Today's Summary

Downloaded the necessary materials for testing (Field testing shortcut, coverage map app). Tested all materials that we added. Added the screenshots and completed the Google Docs. Finished the ReadMe in the GitHub with all the necessary details for the project. Did some research on what results to actually look at while testing, and what they mean. Started on the Powerpoint presentation.

Daily Summary of Team Activities

Audreane:
I filled in the Title page for the powerpoint, formatted the Google Docs document, looked up the plan information for the iPhone, and did the hardware part of the Google Docs. Did some testing for the field Testing, Internet speed testing, and the coverage map testing. Added the Agenda part in the Google Docs.

Kiara:
I created the GitHub repository. I added Audreane to it and made it public so you can access it. I created the READ.md, LICENSE, slides/, src/, config/, docs/report/, docs/plan/, and verification/. I started and shared the Google slides, and the same for the Google Doc. I mainly focused on the software and hosting part from this doc , then tested the field test, internet speed test,and coverage map and documented it.

Agenda

We will visit the places where we believe the cellular service will have a hard time reaching us. For our first baseline, we will be analyzing the outside of Vanier College, where there are fewer physical obstacles that can cause interference. We will create a second baseline by testing in a normal indoor area of the school. This will allow us to see how simply being inside the building affects the connection because of obstacles such as concrete walls. This will give us a more in-depth understanding of cellular signal strength and allows us to compare our baseline results with our hypothesis that the network should be weaker in the bathrooms, basement, and gym. At each location, we will use Field Test Mode to record the cellular signal strength. We will also perform an Internet speed test and record the download speed, upload speed, and latency. We will also use the Coverage map and compare the results to it. We will repeat the same process using both 5G and LTE so that we can compare how they perform in the same location.